



Project Completed: Multi-Environment Infrastructure Design using Terraform Workspaces

Project Overview

In this project, I implemented a multi-environment AWS infrastructure using a single Terraform codebase. The goal was to separate Development, Staging, and Production environments while ensuring infrastructure consistency, scalability, and code reusability.

Key Steps

- Created Terraform Workspaces for Dev, Staging, and Production
- Provisioned AWS EC2 instances for each environment
- Configured Security Groups with environment-specific rules
- Created and managed AWS S3 bucket resources
- Used variables and reusable Terraform modules
- Implemented remote state management using S3 backend
- Validated environment isolation and resource separation

Technologies Used

Terraform | AWS EC2 | AWS S3 | VPC | Security Groups | Terraform Workspaces | Infrastructure as Code (IaC)

step-by-step setup Guide

STEP 1: first of all i had create ec2 instance



```
ec2-user@ip-172-31-16-46:~/terraform-project
"Account": "799884780973",
"Arn": "arn:aws:iam::799884780973:user/terraform-user"
}
[ec2-user@ip-172-31-16-46 terraform-project]$ ls
main.tf  outputs.tf  terraform.tfstate  variables.tf
[ec2-user@ip-172-31-16-46 terraform-project]$ cat main.tf
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = "ap-south-1"
}

resource "aws_s3_bucket" "test_bucket" {
  bucket = "pranita-terraform-demo-12345"

  tags = {
    Name = "TerraformBucket"
  }
}
[ec2-user@ip-172-31-16-46 terraform-project]$ |
```

Step5: aws s3 bucket

The screenshot shows the AWS S3 console interface. The left sidebar contains navigation links for Buckets, Files, Access management and security, and Storage management and insights. The main content area is titled 'Buckets' and shows a list of 'General purpose buckets (1)'. The bucket 'pranita-terraform-demo-12345' is listed with the AWS Region 'Asia Pacific (Mumbai) ap-south-1' and a creation date of 'June 21, 2026, 09:37:37 (UTC-07:00)'. Below the bucket list, there are sections for 'Account snapshot' and 'External access summary'.

STEP6: terraform init

```
ec2-user@ip-172-31-16-46:~/terraform-project

tags = {
  Name = "TerraformBucket"
}
}
[ec2-user@ip-172-31-16-46 terraform-project]$ terraform init
Initializing provider plugins found in the configuration...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v5.100.0

Initializing the backend...

Initializing provider plugins found in the state...
- Reusing previous version of hashicorp/aws
- Using previously-installed hashicorp/aws v5.100.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
[ec2-user@ip-172-31-16-46 terraform-project]$
```

STEP7:terraform apply

```
ec2-user@ip-172-31-16-46:~/terraform-project

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
[ec2-user@ip-172-31-16-46 terraform-project]$ terraform plan
aws_s3_bucket.test_bucket: Refreshing state... [id=pranita-terraform-demo-12345]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no
differences, so no changes are needed.
[ec2-user@ip-172-31-16-46 terraform-project]$ terraform apply
aws_s3_bucket.test_bucket: Refreshing state... [id=pranita-terraform-demo-12345]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no
differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
[ec2-user@ip-172-31-16-46 terraform-project]$
```

Project Outcome

This project helped me understand Infrastructure as Code, Terraform Workspaces, environment isolation, reusable modules, and AWS resource management. It also provided hands-on experience with managing multiple environments from a single codebase.



Key Learning:

Managing Dev, Staging, and Production environments efficiently using Terraform while maintaining security, scalability, and code reusability.

#Terraform #AWS #DevOps #CloudComputing #InfrastructureAsCode #TerraformWorkspaces #EC2 #S3